Vectors I Basics Worksheet: Solutions

Compiled by Maximus Pung Jun

Basics of vectors: **i**, **j**, **k** notation, magnitude, unit vectors, parallel vectors, collinear points, the ratio theorem, the scalar product and the vector product.

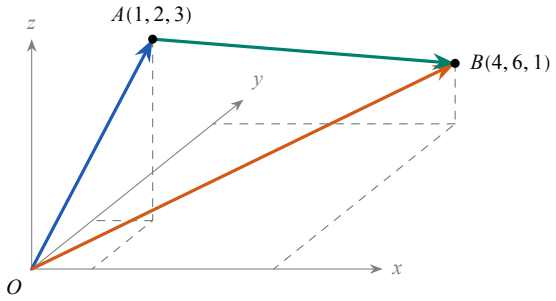
1 **i**, **j**, **k** Notation and Column Vectors

1 Write $\mathbf{a}=3\mathbf{i}-2\mathbf{j}+5\mathbf{k}$ as a column vector, and write $\mathbf{b}=\begin{pmatrix}4\\0\\-1\end{pmatrix}$ in terms of **i**, **j** and **k**.

$$\mathbf{a}=\begin{pmatrix}3\\-2\\5\end{pmatrix}$$

$$\begin{aligned}\mathbf{b}&=4\mathbf{i}+0\mathbf{j}-1\mathbf{k}\\&=4\mathbf{i}-\mathbf{k}\end{aligned}$$

2 The points A and B have coordinates $(1,2,3)$ and $(4,6,1)$. Find $\overrightarrow{AB}$ in terms of **i**, **j** and **k**.



$$\begin{aligned}\overrightarrow{AB}&=\overrightarrow{OB}-\overrightarrow{OA}\\&=\begin{pmatrix}4\\6\\1\end{pmatrix}-\begin{pmatrix}1\\2\\3\end{pmatrix}\\&=\begin{pmatrix}3\\4\\-2\end{pmatrix}\\&=3\mathbf{i}+4\mathbf{j}-2\mathbf{k}\end{aligned}$$

3 Given $\mathbf{p}=2\mathbf{i}+\mathbf{k}$ and $\mathbf{q}=-\mathbf{i}+3\mathbf{j}$, find **r** such that $2\mathbf{p}-3\mathbf{q}+\mathbf{r}=\mathbf{0}$. Give **r** in terms of **i**, **j** and **k**.

$$\mathbf{p}=\begin{pmatrix}2\\0\\1\end{pmatrix},\quad \mathbf{q}=\begin{pmatrix}-1\\3\\0\end{pmatrix}$$

$$\begin{aligned}\mathbf{r}&=3\mathbf{q}-2\mathbf{p}\\&=3\begin{pmatrix}-1\\3\\0\end{pmatrix}-2\begin{pmatrix}2\\0\\1\end{pmatrix}\\&=\begin{pmatrix}-3\\9\\0\end{pmatrix}-\begin{pmatrix}4\\0\\2\end{pmatrix}\\&=\begin{pmatrix}-7\\9\\-2\end{pmatrix}\\&=-7\mathbf{i}+9\mathbf{j}-2\mathbf{k}\end{aligned}$$

4 For the following statements, state whether it is **true** or **false**.

(a) $\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$ **True**

$\mathbf{i}+\mathbf{j}+\mathbf{k}=\begin{pmatrix}1\\1\\1\end{pmatrix}$

(b) The **j**-component of $4\mathbf{i}-\mathbf{j}+2\mathbf{k}$ is -1 **True**

$4\mathbf{i}-\mathbf{j}+2\mathbf{k}=\begin{pmatrix}4\\-1\\2\end{pmatrix}$

(c) The position vector of $P(1,-2,0)$ is $\mathbf{i}-2\mathbf{j}$ **True**

$\overrightarrow{OP}=\begin{pmatrix}1\\-2\\0\end{pmatrix}=\mathbf{i}-2\mathbf{j}$

(d) $3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\2\\0\end{pmatrix}$ **False**

$3\mathbf{i}+2\mathbf{k}=\begin{pmatrix}3\\0\\2\end{pmatrix}\neq\begin{pmatrix}3\\2\\0\end{pmatrix}$

(e) $\overrightarrow{AB}=\overrightarrow{OA}-\overrightarrow{OB}$ **False**

$\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}$

2 Magnitude

1 Find $|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}|$.

$$\begin{aligned}|2\mathbf{i}-3\mathbf{j}+6\mathbf{k}| &= \sqrt{2^2+(-3)^2+6^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

2 The points A and B have coordinates $(1, 2, -1)$ and $(3, -4, 2)$. Find the distance AB.

$$\begin{aligned}\overrightarrow{AB} &= \begin{pmatrix}3\\-4\\2\end{pmatrix}-\begin{pmatrix}1\\2\\-1\end{pmatrix} \\ &= \begin{pmatrix}2\\-6\\3\end{pmatrix}\end{aligned}$$

$$\begin{aligned}AB = \left|\overrightarrow{AB}\right| &= \sqrt{2^2+(-6)^2+3^2} \\ &= \sqrt{49} \\ &= 7\end{aligned}$$

3 Given $\mathbf{a}=\begin{pmatrix}t\\2\\t-1\end{pmatrix}$ and $|\mathbf{a}|=3$, find the possible values of t.

$$\begin{aligned}|\mathbf{a}|^2 &= t^2+2^2+(t-1)^2 \\ 9 &= t^2+4+t^2-2t+1 \\ 0 &= 2t^2-2t-4 \\ 0 &= t^2-t-2 \\ 0 &= (t-2)(t+1) \\ t &= 2 \text{ or } t=-1\end{aligned}$$

4 For the following statements, state whether it is **true** or **false**.

(a) $|-\mathbf{a}|=|\mathbf{a}|$ for every vector $\mathbf{a}$ **True**

$|-\mathbf{a}|=|-1|\,|\mathbf{a}|=|\mathbf{a}|$

(b) $|3\mathbf{a}|=3|\mathbf{a}|$ **True**

$|3\mathbf{a}|=|3|\,|\mathbf{a}|=3|\mathbf{a}|$

(c) $\left|\begin{pmatrix}0\\-5\\0\end{pmatrix}\right|=-5$ **False**

$\left|\begin{pmatrix}0\\-5\\0\end{pmatrix}\right|=\sqrt{0^2+(-5)^2+0^2}=5$

(d) $\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right|=1$ **False**

$\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right|=\sqrt{1^2+1^2+1^2}=\sqrt{3}\neq 1$

(e) $|\mathbf{i}+2\mathbf{j}-2\mathbf{k}|=1+2-2=1$ **False**

$|\mathbf{i}+2\mathbf{j}-2\mathbf{k}|=\sqrt{1^2+2^2+(-2)^2}=3$

(f) $|\mathbf{a}+\mathbf{b}|=|\mathbf{a}|+|\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$ **False**

$|\mathbf{i}+\mathbf{j}|=\sqrt{2}\neq|\mathbf{i}|+|\mathbf{j}|=2$

(g) $|\mathbf{a}+\mathbf{b}|\leq|\mathbf{a}|+|\mathbf{b}|$ for all vectors $\mathbf{a}$, $\mathbf{b}$ **True**

$\overrightarrow{OA}=\mathbf{a}$, $\overrightarrow{AB}=\mathbf{b}$, $\overrightarrow{OB}=\mathbf{a}+\mathbf{b}$, $OB\leq OA+AB$ ($\triangle OAB$)

(h) For all $\lambda\in\mathbb{R}$, $|\lambda\mathbf{a}|=\lambda|\mathbf{a}|$ **False**

$\lambda=-1$, $\mathbf{a}=\mathbf{i}$: $|-\mathbf{i}|=1\neq-1$; $|\lambda\mathbf{a}|=|\lambda||\mathbf{a}|$

3 Unit Vectors

1 Find the unit vector in the direction of $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

$$\left|\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}\right| = \sqrt{2^2+(-1)^2+2^2}$$
$$= 3$$
$$\text{unit vector} = \frac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$$

2 Find the vector of magnitude 10 in the direction of $\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$.

$$\left|\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}\right| = \sqrt{3^2+0^2+(-4)^2}$$
$$= 5$$
$$\text{vector} = 10 \cdot \frac{1}{5}\begin{pmatrix} 3 \\ 0 \\ -4 \end{pmatrix}$$
$$= \begin{pmatrix} 6 \\ 0 \\ -8 \end{pmatrix}$$

3 The points A and B have coordinates $(1, 0, 2)$ and $(3, -6, 5)$. Find the vector of magnitude 14 in the direction *opposite* to $\overrightarrow{AB}$.

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ -6 \\ 5 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$\left|\overrightarrow{AB}\right| = \sqrt{2^2+(-6)^2+3^2}$$
$$= 7$$
$$\text{vector} = -14 \cdot \frac{1}{7}\begin{pmatrix} 2 \\ -6 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} -4 \\ 12 \\ -6 \end{pmatrix}$$

4 For the following statements, state whether it is **true** or **false**.

(a) $\begin{pmatrix}1\\1\\1\end{pmatrix}$ is a unit vector **False**

$$\left|\begin{pmatrix}1\\1\\1\end{pmatrix}\right| = \sqrt{3} \neq 1$$

(b) $\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}$ is a unit vector **True**

$$\left|\frac{1}{3}\begin{pmatrix}1\\2\\2\end{pmatrix}\right| = \frac{1}{3}\sqrt{1+4+4} = 1$$

(c) $\mathbf{i}+\mathbf{j}$ is a unit vector **False**

$|\mathbf{i}+\mathbf{j}| = \sqrt{1^2+1^2+0^2} = \sqrt{2} \neq 1$

(d) $|\hat{\mathbf{a}}| = 1$ for every non-zero vector $\mathbf{a}$ **True**

$$|\hat{\mathbf{a}}| = \left|\frac{\mathbf{a}}{|\mathbf{a}|}\right| = \frac{|\mathbf{a}|}{|\mathbf{a}|} = 1$$

(e) $\mathbf{a}$ and $\hat{\mathbf{a}}$ are parallel **True**

$\mathbf{a} = |\mathbf{a}|\,\hat{\mathbf{a}}$

4 Parallel Vectors ($\mathbf{a} = \lambda \mathbf{b}$)

1 Show that $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$ are parallel.

$$\mathbf{b} = \begin{pmatrix} -6 \\ 3 \\ -9 \end{pmatrix}$$
$$= -3 \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$
$$= -3\mathbf{a}$$
$$\therefore \mathbf{a} \parallel \mathbf{b} \text{ (shown)}$$

2 Find p and q such that $\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$.

$$\begin{pmatrix} 4 \\ p \\ -2 \end{pmatrix} = \lambda \begin{pmatrix} -2 \\ 3 \\ q \end{pmatrix}$$
$$4 = -2\lambda$$
$$\lambda = -2$$
$$p = 3\lambda$$
$$= -6$$
$$-2 = \lambda q$$
$$-2 = -2q$$
$$q = 1$$

3 Find all vectors of magnitude 6 that are parallel to $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$.

$$|\mathbf{a}| = \sqrt{1^2 + 2^2 + (-2)^2}$$
$$= 3$$
$$\text{vector} = \pm 6 \cdot \frac{1}{3} \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$
$$= \pm \begin{pmatrix} 2 \\ 4 \\ -4 \end{pmatrix}$$

4 For the following statements, state whether it is **true** or **false**.

(a) If $\mathbf{a} \parallel \mathbf{b}$, then $\mathbf{a} = \mathbf{b}$ **False**

$2\mathbf{i} \parallel \mathbf{i}$, $2\mathbf{i} \neq \mathbf{i}$

(b) $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ is parallel to $\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix}$ **True**

$$\begin{pmatrix} -2 \\ 4 \\ -6 \end{pmatrix} = -2 \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$$

(c) $\begin{pmatrix}2\\4\\6\end{pmatrix}$ is parallel to $\begin{pmatrix}1\\2\\4\end{pmatrix}$ **False**

$\frac{2}{1}=\frac{4}{2}\neq\frac{6}{4}$

(d) $\begin{pmatrix}3\\0\\-6\end{pmatrix}$ is parallel to $\begin{pmatrix}-1\\0\\2\end{pmatrix}$ **True**

$\begin{pmatrix}3\\0\\-6\end{pmatrix}=-3\begin{pmatrix}-1\\0\\2\end{pmatrix}$

(e) If $\mathbf{a}=-2\mathbf{b}$, then **a** and **b** are not parallel since they point in opposite directions **False**

$\mathbf{a}=\lambda\mathbf{b},\ \lambda=-2$

5 Collinear Points

1 Show that the points $A(1,2,3)$, $B(3,5,4)$ and $C(7,11,6)$ are collinear.

$$\overrightarrow{AB} = \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$
$$= \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$$
$$\overrightarrow{BC} = \begin{pmatrix} 7 \\ 11 \\ 6 \end{pmatrix} - \begin{pmatrix} 3 \\ 5 \\ 4 \end{pmatrix}$$
$$= \begin{pmatrix} 4 \\ 6 \\ 2 \end{pmatrix}$$
$$= 2\overrightarrow{AB}$$
$$\therefore \overrightarrow{AB} \parallel \overrightarrow{BC},\ B \text{ common}$$
$$\therefore A, B, C \text{ collinear (shown)}$$

2 For the points in Question 1, find the ratio $AB : BC$.

$$\overrightarrow{BC} = 2\overrightarrow{AB}$$
$$BC = 2AB$$
$$AB : BC = 1 : 2$$

3 The points $P(2,-1,0)$, $Q(4,1,3)$ and $R(a,7,b)$ are collinear. Find a and b.

$$\overrightarrow{PR} = \lambda \overrightarrow{PQ}$$
$$\begin{pmatrix} a-2 \\ 8 \\ b \end{pmatrix} = \lambda \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$
$$8 = 2\lambda$$
$$\lambda = 4$$
$$a - 2 = 2(4)$$
$$a = 10$$
$$b = 3(4)$$
$$= 12$$

4 For the following statements, state whether it is **true** or **false**.

(a) If $\overrightarrow{AB} \parallel \overrightarrow{CD}$, then A, B, C and D are collinear **False**

$A(0,0,0), B(1,0,0), C(0,1,0), D(1,1,0) : \overrightarrow{AB} = \overrightarrow{CD}$

(b) If $\overrightarrow{AB} = 3\overrightarrow{AC}$, then A, B and C are collinear **True**

$\overrightarrow{AB} \parallel \overrightarrow{AC},\ A$ common

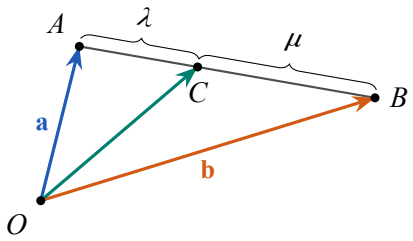
(c) $A(0,0,0)$, $B(1,1,1)$ and $C(2,2,3)$ are collinear **False**

$$\overrightarrow{AC} = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix} \neq \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

(d) The points with position vectors $\mathbf{a}$, $2\mathbf{a}$ and $-\mathbf{a}$ are collinear ($\mathbf{a} \neq \mathbf{0}$) **True**

$2\mathbf{a} - \mathbf{a} = \mathbf{a},\ -\mathbf{a} - \mathbf{a} = -2(\mathbf{a})$

6 Ratio Theorem



In the True/False statements, C lies on AB with $AC:CB=\lambda:\mu$, $\overrightarrow{OA}=\mathbf{a}$ and $\overrightarrow{OB}=\mathbf{b}$.

1 Let $\overrightarrow{OA}=\mathbf{i}+\mathbf{j}+\mathbf{k}$ and $\overrightarrow{OB}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$. Given that C lies on AB, find the coordinates of C if

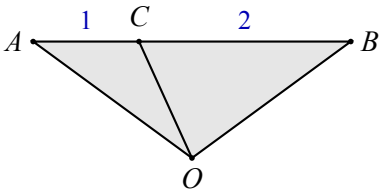
(a) $AC:CB=1:2$,

(b) $3AC=2CB$,

(c) $2AB=5CB$,

(d) Given instead that C lies on AB produced, and $3AC=7BC$.

(a)

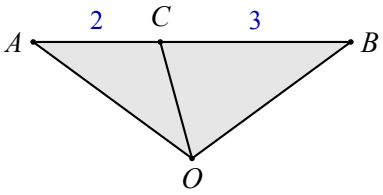


$$\overrightarrow{OC}=\frac{2\overrightarrow{OA}+\overrightarrow{OB}}{3}$$

$$=\begin{pmatrix}1\\ \frac{4}{3}\\ \frac{5}{3}\end{pmatrix}$$

$$\therefore C\left(1,\tfrac{4}{3},\tfrac{5}{3}\right)$$

(b)



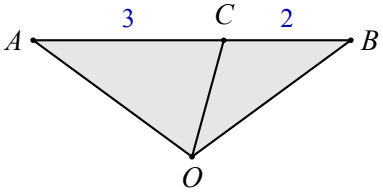
$$\frac{AC}{CB}=\frac{2}{3}$$

$$\overrightarrow{OC}=\frac{2\overrightarrow{OB}+3\overrightarrow{OA}}{5}$$

$$=\begin{pmatrix}1\\ \frac{7}{5}\\ \frac{9}{5}\end{pmatrix}$$

$$\therefore C\left(1,\tfrac{7}{5},\tfrac{9}{5}\right)$$

(c)



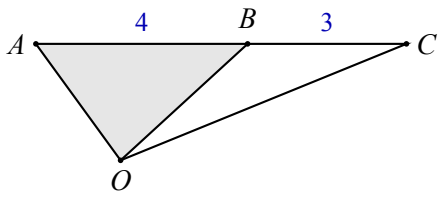
$$\frac{AB}{CB}=\frac{5}{2}$$

$$\overrightarrow{OC}=\frac{3\overrightarrow{OB}+2\overrightarrow{OA}}{5}$$

$$=\begin{pmatrix}1\\ \frac{8}{5}\\ \frac{11}{5}\end{pmatrix}$$

$$\therefore C\left(1,\tfrac{8}{5},\tfrac{11}{5}\right)$$

(d)



$$\frac{AC}{BC}=\frac{7}{3}$$
$$\overrightarrow{OB}=\frac{3\overrightarrow{OA}+4\overrightarrow{OC}}{7}$$
$$\overrightarrow{OC}=\frac{7\overrightarrow{OB}-3\overrightarrow{OA}}{4}$$
$$=\begin{pmatrix}1\\ \frac{11}{4}\\ \frac{9}{2}\end{pmatrix}$$
$$\therefore C\left(1,\tfrac{11}{4},\tfrac{9}{2}\right)$$

2 Find the position vector of the midpoint M of AB, where A and B have coordinates $(3,-1,5)$ and $(-1,5,1)$.

$$\overrightarrow{OM}=\frac{1}{2}\left[\begin{pmatrix}3\\-1\\5\end{pmatrix}+\begin{pmatrix}-1\\5\\1\end{pmatrix}\right]$$
$$=\frac{1}{2}\begin{pmatrix}2\\4\\6\end{pmatrix}$$
$$=\begin{pmatrix}1\\2\\3\end{pmatrix}$$

3 For the following statements, state whether it is **true** or **false**.

(a) $\overrightarrow{OC}=\dfrac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$ **True**

$\overrightarrow{OC}=\mathbf{a}+\frac{\lambda}{\lambda+\mu}(\mathbf{b}-\mathbf{a})=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(b) $\overrightarrow{OC}=\dfrac{\lambda\mathbf{a}+\mu\mathbf{b}}{\lambda+\mu}$ **False**

$\overrightarrow{OC}=\frac{\mu\mathbf{a}+\lambda\mathbf{b}}{\lambda+\mu}$

(c) If $\lambda:\mu=1:1$, then $\overrightarrow{OC}=\dfrac{1}{2}(\mathbf{a}+\mathbf{b})$ **True**

$\overrightarrow{OC}=\frac{1\mathbf{a}+1\mathbf{b}}{2}$

(d) If $\lambda:\mu=1:3$, then $\overrightarrow{OC}=\dfrac{3\mathbf{a}+\mathbf{b}}{4}$ **True**

$\overrightarrow{OC}=\frac{3\mathbf{a}+1\mathbf{b}}{1+3}$

(e) If $\lambda:\mu=2:1$, then $\overrightarrow{OC}=\dfrac{2\mathbf{a}+\mathbf{b}}{3}$ **False**

$\overrightarrow{OC}=\frac{1\mathbf{a}+2\mathbf{b}}{3}$

7 Scalar Product

1 Properties of scalar product

It is given that $\mathbf{a}=\begin{pmatrix}1\\2\\3\end{pmatrix}$, $\mathbf{b}=\begin{pmatrix}-3\\2\\7\end{pmatrix}$, $\mathbf{c}=\begin{pmatrix}4\\-5\\-1\end{pmatrix}$. Find the following

(a) $\mathbf{a}\cdot\mathbf{b}$ (b) $\mathbf{b}\cdot\mathbf{a}$ (c) $\mathbf{a}\cdot(\mathbf{b}+\mathbf{c})$ (d) $\mathbf{a}\cdot\mathbf{b}+\mathbf{a}\cdot\mathbf{c}$

(e) $\mathbf{a}\cdot(2\mathbf{c})$ (f) $(2\mathbf{a})\cdot\mathbf{c}$ (g) $\mathbf{a}\cdot\mathbf{a}$

Ans: (a) 22 (b) 22 (c) 13 (d) 13 (e) -18 (f) -18 (g) 14

2 Scalar product formula: $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$

(a) Points O, A, B are such that $\overrightarrow{OA}=\mathbf{i}+4\mathbf{j}-3\mathbf{k}$ and $\overrightarrow{OB}=5\mathbf{i}-\mathbf{j}$

(i) Find the size of angle AOB

(ii) Find the size of angle OAB.

(b) It is given that $\mathbf{c}=\begin{pmatrix}2\\3\\6\end{pmatrix}$, $\mathbf{d}=\begin{pmatrix}1\\-2\\3\end{pmatrix}$ and $\mathbf{e}=\begin{pmatrix}3\\2\\-2\end{pmatrix}$. Find the angle between the vectors $\mathbf{c}$ and

(i) the vector $\mathbf{d}$

(ii) the vector $\mathbf{e}$

(iii) the positive y-axis

(c) It is given that two vectors $\mathbf{a}$ and $\mathbf{b}$ satisfy the following relationship:

$$|\mathbf{a}||\mathbf{b}|=6.$$

Find the value of $\mathbf{a}\cdot\mathbf{b}$ for each of the following scenarios.

(i) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $60°$.

(ii) The angle between $\mathbf{a}$ and $\mathbf{b}$ is $120°$.

(iii) Vectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Ans:

(a)(i) $87.8°$ (ii) $133.9°$

(b)(i) $57.7°$ (ii) $90°$ (iii) $64.6°$

(c)(i) $\mathbf{a}\cdot\mathbf{b}=3$ (ii) $\mathbf{a}\cdot\mathbf{b}=-3$ (iii) $\mathbf{a}\cdot\mathbf{b}=0$

3 Perpendicularity of vectors

(a) Referred to the origin O, the position vectors of the points A and B are $\mathbf{i}-\mathbf{j}+2\mathbf{k}$ and $2\mathbf{i}+4\mathbf{j}+\mathbf{k}$ respectively. Show that OA is perpendicular to OB.

(b) The position vectors $\mathbf{a}$ and $\mathbf{b}$ are given by

$$\mathbf{a}=2p\mathbf{i}+3p\mathbf{j}+6p\mathbf{k} \quad \text{and} \quad \mathbf{b}=\mathbf{i}-2\mathbf{j}+2\mathbf{k}$$

where $p>0$. It is given that $|\mathbf{a}|=|\mathbf{b}|$.

(i) Find the exact value of p.

(ii) Show that the vectors $(\mathbf{a}+\mathbf{b})$ and $(\mathbf{a}-\mathbf{b})$ are perpendicular.

Ans: (b)(i) $p=\frac{3}{7}$

4 Length of projection

(a) Given that $\mathbf{a}=2\mathbf{i}+\mathbf{j}+2\mathbf{k}$ and $\mathbf{b}=-\mathbf{i}+2\mathbf{k}-\mathbf{k}$, find the length of projection and the projection vector of
   (i) **a** onto **b**,
   (ii) **b** onto **a**.

(b) With reference to the origin O, the points A, B and C have position vectors given by

$$\mathbf{a}=\mathbf{i}+2\mathbf{j}-\mathbf{k},\ \mathbf{b}=3\mathbf{i}+\mathbf{k} \text{ and } \mathbf{c}=2\mathbf{i}+2\mathbf{j}+3\mathbf{k}$$

respectively. Find the length of projection of $\overrightarrow{BA}$ onto $\overrightarrow{BC}$.

Ans:

(a)(i) $\frac{2}{\sqrt{6}}, \frac{1}{3}\begin{pmatrix}-1\\2\\-1\end{pmatrix}$ (ii) $\frac{2}{3}, \frac{2}{9}\begin{pmatrix}2\\1\\2\end{pmatrix}$

(b) $\frac{2}{3}$

5 For the following statements, state whether it is **true** or **false**.

(a) $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$. **True**

$a_1b_1+a_2b_2+a_3b_3=b_1a_1+b_2a_2+b_3a_3$

(b) $\mathbf{a}\cdot\mathbf{a}=|\mathbf{a}|^2$. **True**

$\mathbf{a}\cdot\mathbf{a}=a_1^2+a_2^2+a_3^2=|\mathbf{a}|^2$

(c) $\mathbf{i}\cdot\mathbf{j}=0$. **True**

$1\cdot0+0\cdot1+0\cdot0=0$

(d) $\mathbf{a}\cdot\mathbf{b}$ is a vector. **False**

$\begin{pmatrix}1\\0\\0\end{pmatrix}\cdot\begin{pmatrix}1\\0\\0\end{pmatrix}=1$ (a scalar)

(e) $\mathbf{i}\cdot\mathbf{j}=\mathbf{k}$. **False**

$\mathbf{i}\cdot\mathbf{j}=0$

(f) If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$. **True**

$\mathbf{0}\cdot\mathbf{b}=0b_1+0b_2+0b_3=0$

(g) $(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$. **True**

$(2\mathbf{a})\cdot(3\mathbf{b})=2\cdot3\ (\mathbf{a}\cdot\mathbf{b})$

(h) $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector. **True**

$(\mathbf{a}\cdot\mathbf{b})\mathbf{c}=\lambda\mathbf{c},\ \lambda=\mathbf{a}\cdot\mathbf{b}\in\mathbb{R}$

(i) $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$, where θ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$. **True**

Definition of scalar product

(j) $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$. **False**

$\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=2+6+15=23$ (a scalar)

(k) If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. **False**

$\mathbf{i}\cdot\mathbf{j}=0,\ \mathbf{i}\neq\mathbf{0},\ \mathbf{j}\neq\mathbf{0}$

(l) $|\mathbf{a}\cdot\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$. **False**

$|\mathbf{i}\cdot\mathbf{j}|=0\neq1=|\mathbf{i}||\mathbf{j}|$

(m) $\mathbf{a}\cdot\mathbf{b}\leq|\mathbf{a}||\mathbf{b}|$. **True**

$\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta\leq|\mathbf{a}||\mathbf{b}|$

(n) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute. **True**

$|\mathbf{a}||\mathbf{b}|\cos\theta>0\ \therefore\ \cos\theta>0,\ 0^\circ\leq\theta<90^\circ$

(o) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is obtuse. **True**

$\cos\theta<0 \therefore 90^\circ<\theta\leq 180^\circ$

(p) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}\geq 0$ means that the angle between $\mathbf{a}$ and $\mathbf{b}$ is acute. **False**

$\mathbf{i}\cdot\mathbf{j}=0,\ \theta=90^\circ$

(q) $\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar. **False**

$\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore \mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is meaningless

(r) $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=|\mathbf{a}|^2+|\mathbf{b}|^2$. **False**

$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=|\mathbf{a}|^2+2\mathbf{a}\cdot\mathbf{b}+|\mathbf{b}|^2;\ \mathbf{a}=\mathbf{b}=\mathbf{i}: 4\neq 2$

(s) For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel. **True**

$\cos\theta=1 \therefore \theta=0^\circ$

(t) $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\theta$ also holds if θ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$. **True**

$\cos(360^\circ-\theta)=\cos\theta$

(u) Let θ be the angle ABC. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=|\overrightarrow{BA}||\overrightarrow{BC}|\cos\theta$. **True**

$\overrightarrow{BA}$, $\overrightarrow{BC}$ both start at B

(v) The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$. **False**

$\mathbf{a}=\mathbf{i},\ \mathbf{b}=2\mathbf{i}: |\mathbf{a}\cdot\hat{\mathbf{b}}|=1\neq 2=|\mathbf{b}\cdot\hat{\mathbf{a}}|$

(w) If $|\mathbf{a}|=|\mathbf{b}|$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$. **True**

$(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=|\mathbf{a}|^2-|\mathbf{b}|^2=0$

(x) $\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. **False**

$\mathbf{a}=\mathbf{i},\ \mathbf{b}=\mathbf{j},\ \mathbf{c}=\mathbf{k}: \mathbf{a}\cdot\mathbf{b}=0=\mathbf{a}\cdot\mathbf{c},\ \mathbf{b}\neq\mathbf{c}$

(y) Let θ be the angle ABC. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=|\overrightarrow{AB}||\overrightarrow{BC}|\cos\theta$. **False**

$\overrightarrow{AB}\cdot\overrightarrow{BC}=-\overrightarrow{BA}\cdot\overrightarrow{BC}=-|\overrightarrow{AB}||\overrightarrow{BC}|\cos\theta$

(z) Let θ be the angle CBD. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=|\overrightarrow{BD}||\overrightarrow{CB}|\cos\theta$. **False**

$\overrightarrow{BD}\cdot\overrightarrow{CB}=-\overrightarrow{BD}\cdot\overrightarrow{BC}=-|\overrightarrow{BD}||\overrightarrow{CB}|\cos\theta$

8 Vector Product

1 Properties of vector product

(a) Find $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 5 \\ 7 \\ 1 \end{pmatrix}$

(b) Given that $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$, find

(i) $\mathbf{a} \times \mathbf{b}$; (ii) $\mathbf{b} \times \mathbf{a}$; (iii) $\mathbf{a} \times (\mathbf{b} + \mathbf{c})$; (iv) $\mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$;

(v) $\mathbf{a} \times (2\mathbf{b})$; (vi) $(2\mathbf{a}) \times \mathbf{b}$; (vii) $\mathbf{a} \times \mathbf{a}$;

Ans:

(a) $\begin{pmatrix} -29 \\ 17 \\ 26 \end{pmatrix}$

(b)(i) $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$ (ii) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ (iii) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ (iv) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$ (v) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ (vi) $\begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}$ (vii) $\mathbf{0}$

2 Parallel vectors

(a) Three non-zero vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy $\mathbf{a} \times 3\mathbf{b} = 2\mathbf{a} \times \mathbf{c}$. Show that $\mathbf{a}$ and $3\mathbf{b} - 2\mathbf{c}$ are parallel.

(b) Find the scalar product and vector product of the following vectors, stating whether they are parallel, perpendicular, or neither.

(i) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$

(ii) $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, $\mathbf{b} = 3\mathbf{i} + \mathbf{k}$

(iii) $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ 4 \end{pmatrix}$

Ans: (b) Neither; Perpendicular; Parallel

3 Vector product formula

Using the Vector Product, find the acute angles between the following vectors:

(a) $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$ (b) $\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$ (c) $\begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

Ans: (a) 51.0° (b) 46.4° (c) 87.0°

4 Perpendicular distance

(a) Relative to an origin O, two point A and B have position vectors given by $\mathbf{a} = 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = -2\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$. Find the following:

(i) The shortest distance from A to OB

(ii) The shortest distance from B to OA

(b) Relative to the origin O, two points A and B have position vectors given by $\mathbf{a} = \mathbf{i} + \mathbf{j} + \mathbf{k}$ and $\mathbf{b} = \mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ respectively.

(i) Find the length of projection of $\mathbf{b}$ onto $\mathbf{a}$.

(ii) Hence, or otherwise, find the shortest distance from B to OA.

(c) Relative to the origin O, the points A, B and C have position vectors given by $\mathbf{a}=\mathbf{i}+2\mathbf{j}+3\mathbf{k}$, $\mathbf{b}=\mathbf{i}-\mathbf{j}+\mathbf{k}$ and $\mathbf{c}=3\mathbf{i}-\mathbf{j}+\mathbf{k}$. Find the shortest distance from C to the line joining A and B.

Ans:

(a)(i) 4.09 units (ii) 4.51 units

(b)(i) $2\sqrt{3}$ units (ii) $\sqrt{2}$ units

(c) 2 units

5 Areas of triangles & parallelograms

(a) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA}=\begin{pmatrix}1\\0\\3\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}-2\\3\\2\end{pmatrix}$. Find the area of triangle OAB.

(b) Relative to the origin O, the points A and B have position vectors $\overrightarrow{OA}=\begin{pmatrix}0\\1\\1\end{pmatrix}$, $\overrightarrow{OB}=\begin{pmatrix}1\\-2\\2\end{pmatrix}$. Find the area of triangle OAB.

(c) Relative to an origin O, the position vectors of the vertices A, B and C of a triangle are

$$\mathbf{i}+\mathbf{j}-\mathbf{k},\ 2\mathbf{i}+\mathbf{k} \text{ and } -\mathbf{i}+2\mathbf{j}-\mathbf{k}$$

respectively. By considering the displacement vectors $\overrightarrow{AB}$ and $\overrightarrow{AC}$, find the exact area of triangle ABC.

Ans: (a) $\frac{\sqrt{154}}{2}$ units^2 (b) $\frac{3\sqrt{2}}{2}$ units^2 (c) $\frac{\sqrt{21}}{2}$ units^2

6 For the following statements, state whether it is **true** or **false**.

(a) $\mathbf{a}\times\mathbf{a}=\mathbf{0}$. **True**

$|\mathbf{a}\times\mathbf{a}|=|\mathbf{a}|^2\sin 0^\circ=0$

(b) $\mathbf{a}\times\mathbf{a}=0$. **False**

$\mathbf{a}\times\mathbf{a}=\mathbf{0}\neq 0$ (left side is the zero vector)

(c) $\mathbf{i}\times\mathbf{j}=\mathbf{k}$. **True**

$$\begin{pmatrix}1\\0\\0\end{pmatrix}\times\begin{pmatrix}0\\1\\0\end{pmatrix}=\begin{pmatrix}0\\0\\1\end{pmatrix}$$

(d) $\mathbf{a}\times\mathbf{b}=\mathbf{b}\times\mathbf{a}$. **False**

$\mathbf{i}\times\mathbf{j}=\mathbf{k},\ \mathbf{j}\times\mathbf{i}=-\mathbf{k}$

(e) $\mathbf{a}\times\mathbf{b}=-(\mathbf{b}\times\mathbf{a})$. **True**

$\mathbf{a}\times\mathbf{b}=-\mathbf{b}\times\mathbf{a}$

(f) $\mathbf{a}\times\mathbf{b}$ is a scalar. **False**

$\mathbf{i}\times\mathbf{j}=\mathbf{k}$ (a vector)

(g) $\mathbf{k}\times\mathbf{j}=\mathbf{i}$. **False**

$\mathbf{j}\times\mathbf{k}=\mathbf{i}\ \therefore\ \mathbf{k}\times\mathbf{j}=-\mathbf{i}$

(h) $\begin{pmatrix}1\\2\\3\end{pmatrix}\times\begin{pmatrix}2\\4\\6\end{pmatrix}=\mathbf{0}$. **True**

$$\begin{pmatrix}2\\4\\6\end{pmatrix}=2\begin{pmatrix}1\\2\\3\end{pmatrix}$$

(i) $\mathbf{a}\times(\mathbf{b}+\mathbf{c})=\mathbf{a}\times\mathbf{b}+\mathbf{a}\times\mathbf{c}$. **True**

Distributive law

(j) $(\lambda\mathbf{a})\times\mathbf{b}=\lambda(\mathbf{a}\times\mathbf{b})$ for all $\lambda\in\mathbb{R}$. **True**

$(\lambda\mathbf{a})\times\mathbf{b}=\mathbf{a}\times(\lambda\mathbf{b})=\lambda(\mathbf{a}\times\mathbf{b})$

(k) $\mathbf{a}\times\mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. **True**

$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{a}=0,\ (\mathbf{a}\times\mathbf{b})\cdot\mathbf{b}=0$

(l) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}$ is a scalar. **True**

$(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\text{vector}\cdot\text{vector}$

(m) $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|\sin\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$. **True**

Definition of vector product

(n) $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$. **False**

$|\mathbf{i}\times\mathbf{i}|=0\neq 1$

(o) If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. **False**

$\mathbf{i}\times 2\mathbf{i}=\mathbf{0},\ \mathbf{i}\neq\mathbf{0},\ 2\mathbf{i}\neq\mathbf{0}$

(p) For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ means that $\mathbf{a}$ is parallel to $\mathbf{b}$. **True**

$|\mathbf{a}||\mathbf{b}|\sin\theta=0\ \therefore\ \theta=0^\circ$ or 180°

(q) The area of the parallelogram with adjacent sides $\mathbf{a}$ and $\mathbf{b}$ is $|\mathbf{a}\times\mathbf{b}|$. **True**

$\text{Area}=|\mathbf{a}||\mathbf{b}|\sin\theta=|\mathbf{a}\times\mathbf{b}|$

(r) The area of triangle OAB is $|\overrightarrow{OA}\times\overrightarrow{OB}|$. **False**

$\text{Area}=\frac{1}{2}|\overrightarrow{OA}\times\overrightarrow{OB}|$

(s) $\mathbf{a}\times(\mathbf{b}\cdot\mathbf{c})$ is a vector. **False**

$\mathbf{b}\cdot\mathbf{c}$ is a scalar $\therefore$ meaningless

(t) For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $|\mathbf{a}\times\mathbf{b}|=|\mathbf{a}||\mathbf{b}|$, then $\mathbf{a}$ is perpendicular to $\mathbf{b}$. **True**

$\sin\theta=1\ \therefore\ \theta=90^\circ$

(u) $(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$. **True**

$\text{both}=\det(\mathbf{a}\ \mathbf{b}\ \mathbf{c})$

(v) $|\mathbf{a}\times\mathbf{b}|^2+(\mathbf{a}\cdot\mathbf{b})^2=|\mathbf{a}|^2|\mathbf{b}|^2$. **True**

$|\mathbf{a}|^2|\mathbf{b}|^2(\sin^2\theta+\cos^2\theta)=|\mathbf{a}|^2|\mathbf{b}|^2$

(w) $\mathbf{a}\times\mathbf{b}=\mathbf{a}\times\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. **False**

$\mathbf{i}\times\mathbf{j}=\mathbf{k}=\mathbf{i}\times(\mathbf{i}+\mathbf{j}),\ \mathbf{j}\neq\mathbf{i}+\mathbf{j}$

(x) $\mathbf{a}\times(\mathbf{b}\times\mathbf{c})=(\mathbf{a}\times\mathbf{b})\times\mathbf{c}$. **False**

$\mathbf{i}\times(\mathbf{i}\times\mathbf{j})=\mathbf{i}\times\mathbf{k}=-\mathbf{j},\ (\mathbf{i}\times\mathbf{i})\times\mathbf{j}=\mathbf{0}$

(y) The shortest distance from B to the line through A parallel to $\mathbf{d}$ is $|\overrightarrow{AB}\times\mathbf{d}|$. **False**

$\text{distance}=|\overrightarrow{AB}\times\hat{\mathbf{d}}|=\dfrac{|\overrightarrow{AB}\times\mathbf{d}|}{|\mathbf{d}|}$

(z) If $\mathbf{a}\times\mathbf{b}=\mathbf{0}$ and $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. **True**

$|\mathbf{a}|^2|\mathbf{b}|^2=|\mathbf{a}\times\mathbf{b}|^2+(\mathbf{a}\cdot\mathbf{b})^2=0$

END OF WORKSHEET